Prashant JAYANNAVAR

PhD | University of Illinois Urbana-Champaign

paj3@illinois.edu · prashant-jayan21.github.io · Google Scholar
github.com/prashant-jayan21 · linkedin.com/in/prashantjayannavar · @p_jayannavar
Thomas M. Siebel Center for Computer Science, 201 North Goodwin Avenue, Urbana, IL 61801

I am a recent CS PhD graduate from UIUC, advised by Prof. Julia Hockenmaier, specializing in **Natural Language Processing (NLP)**. My research centers on instruction-following agents in **embodied dialogue**, using games as simulated environments. Themes include human-AI collaboration, spatial reasoning, SLMs, synthetic data, and evaluation benchmarks.

Previously, I spent 2.5 years as an **Applied Scientist** at the AI startup x.ai (acquired by Bizzabo), building a task-oriented dialogue agent to automate email-based meeting scheduling.

I am actively seeking **Research Scientist** or **Applied Scientist** roles.

EDUCATION

Sept 2017 - Dec 2025 — **PhD in Computer Science**, University of Illinois at Urbana-Champaign, *Urbana, IL*
Advisor: Prof. Julia Hockenmaier (UIUC)
Committee: Prof. Dilek Hakkani-Tur (UIUC), Prof. ChengXiang Zhai (UIUC), Prof. Massimo Poesio (QMUL, UK)
Research Areas: Grounded Instruction Following, Task-Oriented Dialogue, Embodied AI, Datasets and Evaluation, Synthetic Data, Spatial Reasoning in LLMs

Jan 2015 — **MS in Computer Science (Track: Machine Learning)**, Columbia University, *New York, NY*

May 2013 — **B.Tech in Computer Science and Engineering**, National Institute of Technology Rourkela, *Rourkela, India*

RESEARCH & INDUSTRY EXPERIENCE

Dec 2025 – Sept 2017

Research Assistant, UNIVERSITY OF ILLINOIS AT URBANA-CHAMPAIGN, Urbana, IL
Advisor: Prof. Julia Hockenmaier
My research focuses on instruction-following agents and spatial intelligence in embodied dialogue. It develops a novel research ecosystem for this domain by using games as simulated environments:

Part 1: A Novel Testbed
- Created the **Minecraft Dialogue Corpus (MDC)**, a pioneering use of Minecraft in this domain.
- Built and open-sourced the **data collection platform**.
- Formulated the **Builder Action Prediction (BAP)** task for instruction following: evaluation, baselines (CNN+GRU), and data augmentation methods.

Part 2: BAP v2: An Enhanced Task Framework
- Developed an improved **evaluation benchmark** for BAP.
- Designed **novel dialogue simulators** to generate rich **synthetic embodied dialogues**.
- Designed a **Curriculum Learning** strategy to leverage this data and boost spatial reasoning in models.
- Established newer baseline architectures using **Transformers** and early LLMs (**BERT**).

Part 3: Into the Age of Modern LLMs
- Developed the new **state-of-the-art** model, **Llama-CRAFTS**, achieving F1 of 53.0, +6.0 over prior SOTA (**post-training** Llama-3-8B with **SFT**/instruction tuning).
- Demonstrated that BAP v2 provides a key measure of spatial capabilities and brittleness of LLMs.

Grounded Instruction Following | Embodied AI | Spatial Reasoning | Task-Oriented Dialogue | LLMs | Synthetic Data | Evaluation Benchmarks | Data Collection | Minecraft

June 2017 – Feb 2015

Applied Scientist, X.AI (ACQUIRED BY BIZZABO), New York, NY
Built a production **NLP pipeline** for **temporal information extraction** from unstructured email text.
- Developed temporal expression extraction and normalization, combining ML with syntactic parsing and a proprietary temporal markup language.
- Built a dialog act classifier to interpret user intent behind temporal expressions.
- Deployed microservices integrating these capabilities into the main product.
- Implemented online and offline metrics to benchmark and monitor model performance.
- Authored annotation guidelines and supervised annotation, collaborating with annotators, engineers, UI designers, and product leaders.

Temporal Information Extraction | Dialogue Systems | Data Annotation | NLP Engineering | Microservices

Dec 2014 – Jun 2014

Research Assistant, Columbia University, New York, NY
Advisor: Prof. Owen Rambow
Demonstrated the **utility of NLP in digital humanities** research.
- Enhanced the SINNET system for social network extraction and applied it to nineteenth-century British novels to test long-standing literary theories.
- Validated the system by showing high correlation with human annotations and cross-genre generalization from news articles (training data) to literary texts.
- Challenged the prevailing interpretation of the theories, arguing that prior quantitative findings support rather than disprove them, and generated stronger evidence for this revised interpretation.

NLP in Digital Humanities | Social Network Analysis | Computational Literary Analysis

Publications

J=Journal, C=Conference, W=Workshop, D=Demo, R=Report

Selected Papers

[J.1] **BAP v2: An Enhanced Task Framework for Instruction Following in Minecraft Dialogues** [pre-MIT Press publication version | code | blog]
Prashant Jayannavar, Liliang Ren, Marisa Hudspeth, Risham Sidhu, Charlotte Lambert, Ariel Cordes, Elizabeth Kaplan, Anjali Narayan-Chen, Julia Hockenmaier
*Computational Linguistics journal (*selected for oral presentation at ACL 2026)* [**Computational Linguistics; 2025**]

[C.1] **MDC-R: The Minecraft Dialogue Corpus with Reference** [paper]
Chris Madge, Maris Camilleri, Paloma Carretero Garcia, Vanja Karan, Juexi Shao, Prashant Jayannavar, Julian Hough, Benjamin Roth, Massimo Poesio
Under review; arXiv preprint arXiv:2506.22062 [**arXiv; 2025**]

[R.1] **Human in the Loop Learning through Grounded Interaction in Games (Dagstuhl Perspectives Workshop 24492)** [paper | website]
Raffaella Bernardi, Julia Hockenmaier, Udo Kruschwitz, Prashant Jayannavar, Massimo Poesio
Dagstuhl Reports, Vol. 14, Issue 12, Schloss Dagstuhl – Leibniz-Zentrum für Informatik (2025) [**Dagstuhl Reports 2025**]

[C.2] **Learning to execute instructions in a Minecraft dialogue** [paper | video | slides | code | website]
Prashant Jayannavar, Anjali Narayan-Chen, Julia Hockenmaier
58th Annual Meeting of the Association for Computational Linguistics (Virtual) [**ACL 2020**]

[C.3] **Collaborative Dialogue in Minecraft** [paper | data | code (models) | code (platform) | website]
Anjali Narayan-Chen*, Prashant Jayannavar*, Julia Hockenmaier (* = Equal Contribution)
57th Annual Meeting of the Association for Computational Linguistics, Florence, Italy [**ACL 2019**]

Other Papers

[C.4] **Lara - Human-guided collaborative problem solver: Effective integration of learning, reasoning and communication** [paper | supplemental | blog | video]
Harsha Kokel, Mayukh Das, Rakibul Islam, Julia Bonn, Jon Cai, Soham Dan, Anjali Narayan-Chen, Prashant Jayannavar, Janardhan Rao Doppa, Julia Hockenmaier, Sriraam Natarajan, Martha Palmer, Dan Roth
Tenth Annual Conference on Advances in Cognitive Systems, Arlington, Virginia [**ACS 2022**]

[D.1] **Human-guided Collaborative Problem Solving: A Natural Language based Framework** [paper | blog | video]
Harsha Kokel, Mayukh Das, Rakibul Islam, Julia Bonn, Jon Cai, Soham Dan, Anjali Narayan-Chen, Prashant Jayannavar, Janardhan Rao Doppa, Julia Hockenmaier, Sriraam Natarajan, Martha Palmer, Dan Roth
Thirty-First International Conference on Automated Planning and Scheduling, Guangzhou, China [**ICAPS 2021**]

[W.1] **Validating Literary Theories Using Automatic Social Network Extraction** [paper]
Prashant Jayannavar, Apoorv Agarwal, Melody Ju, Owen Rambow
NAACL HLT Fourth Workshop on Computational Linguistics for Literature, Denver, Colorado [**CLFL, NAACL HLT 2015**]

Mentions & Media Coverage

- **Prof. Martha Palmer** (July 2023) — *My Big, Fat 50-Year Journey* (2023 ACL Lifetime Achievement Award acceptance speech). [Transcript in Computational Linguistics Journal]
- **Queen Mary University of London EECS News** (Nov 2021) — *EECS researcher wins over £1million of funding for research project* (featuring Prof. Massimo Poesio). [Article]
- **U.S. Army DEVCOM Army Research Laboratory Public Affairs** (July 2021) — *Army award-winning research to transform Soldier-robot communication.* [Article]

Technical Skills

Programming:	Python, Scala, Java, C++
ML and NLP Libraries:	PyTorch, NumPy, SciPy, scikit-learn, Stanford CoreNLP, NLTK
LLM Engineering:	Hugging Face (Transformers, Datasets, TRL, PEFT, Pipeline), Post-training (SFT), QLoRA, Distributed Training (Accelerate), BitsAndBytes
Data Analysis & Viz:	Pandas, Matplotlib, Plotly, Jupyter Widgets (ipywidgets)
Tools & Platforms:	Git, TensorBoard, Slurm, LaTeX, Unix/Linux, AWS, MongoDB, gRPC, Protobuf

Mentorship

Master's students

- **Amit Athani** (MS CS, UIUC → Palantir) — Master's Thesis (2023).
  Topic: *Introducing Builder Perspective in Minecraft Architect Utterance Generation*. [Thesis]
- **Dhruv Agarwal** (MS CS, UIUC → Amazon) — Master's Thesis (2019).
  Topic: *Multi Task Learning and Incorporating Common Sense Knowledge for Question Answering*. [Thesis]

Undergraduate students

- **Marisa Hudspeth** (BS CS, Rhodes College → PhD, UMass, Amherst) — NSF DREU Research Internship (2021).
  Project: *Synthetic Data Generation and Dialogue Simulation for Embodied AI in Minecraft*.
  Outcome: Co-author on **publication [J.1]**. [DREU Program | Paper]
- **Charlotte Lambert** (BA CS, Vassar College → PhD CS, UIUC), **Ariel Cordes** (BA CS, UMN–Morris), **Elizabeth Kaplan** (BS CS, NCSU → Amazon) — NSF DREU Research Internship (2019).
  Project: *Virtual World Context Encoding for Grounded Dialogue in Minecraft*.
  Outcome: Received the **Best Research Project Award**; Co-author on **publication [J.1]**. [DREU Program | Report | Paper | Award Details]
- **Hetvi Patel** (BS CS, UIUC), **Sana Madhavan** (BS CS+Linguistics, UIUC) — UIUC CS STARS Research Program (2024).
  Project: *Data Annotation for Fairer Evaluation for the Minecraft Builder Task*.
  Outcome: Acknowledged in publication [J.1]. [CS STARS Program]

Teaching Experience

May 2025 – Sept 2021 | **Teaching Assistant, University of Illinois at Urbana- Champaign, Urbana, IL**
Performed TA duties including holding office hours and providing academic support for students in CS 447: Natural Language Processing (taught by Prof. Julia Hockenmaier).

May 2014 – Feb 2014 | **Academic Tutor, Columbia University, New York, NY**
Tutored Columbia student athletes in Intro to CS and Programming in MATLAB (COMS W1005) and Calculus III (MATH V1201) as part of *Columbia's Student-Athlete Enrichment Services program*.